

# The Elements of Statistics

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## 1 Random variable

### 1.1 Representation

$X$  : upper case, a random variable

$x$  : lower case, a value for the random variable  $X$

We can say  $X \sim P()$  or  $X \sim f()$ , where  $P$  is a probabilistic measure, and  $f$  is a p.d.f (probability density function). But we don't say  $x \sim P()$  or  $x \sim f()$  since  $x$  is a fixed value.

We usually say  $P(X=x) = \frac{1}{2}$  or  $f(x) \propto e^{-x^2}$ .

$\mathbf{X}$ : bold upper case, a matrix or a vector

$\mathbf{x}$ : bold lower case, a vector.

When we say a vector, without explanation, it means a column-wise vector, i.e.,

$$\mathbf{x} = (x_1, x_2, \dots, x_n)^T.$$

### 1.2 Conditional probability

$X|Y=y \sim p(x|y)$  is a conditional probability density function. It's a p.d.f for the random variable  $X$  given a fixed value of  $Y=y$ .

$p(x|y) = \frac{p(x,y)}{p(y)}$ , the definition of conditional p.d.f, where  $p(x,y)$  is the joint p.d.f for the random variables  $X$  and  $Y$ .

$$p(x|y) \cdot p(y) = p(x,y) = p(x) \cdot p(y|x).$$

**Law of total probability:**  $p(y) = \int p(y|x)p(x) dx$ .

**Independence:**  $p(x,y) = p(x) \cdot p(y)$

### 1.3 Expectation

$E(X)$ : expectation of the random variable  $X$ ,  $E(X) = \int x f(x) dx$ .

$$E(g(X)) = \int g(x) f(x) dx$$

$E(X|Y)$ : conditional expectation of the random variable  $X$  given  $Y$ .

$E(X|Y=y) = \int x f(x|y) dx$ , it's a function of  $Y$ . If  $Y$  is a random variable,  $E(X|Y)$  is usually another random variable.

$$E(E(X|Y)) = \int \left[ \int x f(x|y) dx \right] f(y) dy = \int x f(x, y) dx dy = E(X).$$

$E_Y f(X, Y)$ : expectation of  $f(X, Y)$  for the random variable  $Y$ .

Linear operator:  $E(aX + b) = aE(X) + b$

We usually use  $\mu_X$  to represent  $E(X)$ .

Let  $\mathbf{x} = (x_1, \dots, x_n)$  represent  $n$  samples. Then the sample-version expectation is defined as:

$$\tilde{E}(X) = \sum_{i=1}^n x_i / n = \mathbf{1} \cdot \mathbf{x}^T / n. \text{ Here } \mathbf{1} = (1, \dots, 1) \text{ with size of } n.$$

## 1.4 Variance

$$\text{Var}(X) = \int (x - E(X))^2 p(x) dx = EX^2 - (EX)^2$$

$\text{Var}(X|Y = y) = \int (x - E(X|Y = y))^2 p(x|y) dx$ , is a function of  $y$ , and usually a random variable if  $Y$  is a random variable.

$$\text{Var}(X|Y) = E(X^2|Y) - (E(X|Y))^2.$$

$$\text{Var}(X) = \text{Var}(E(X|Y)) + E(\text{Var}(X|Y)).$$

$$\text{Var}(aX + b) = \text{Var}(aX) = a^2 \text{Var}(X)$$

We usually use  $\sigma_X^2$  to represent  $\text{Var}(X)$ , and  $\sigma_X = \sqrt{\sigma_X^2}$  to represent standard deviation of  $X$ .

## 1.5 Covariance

$$\text{Cov}(X, Y) = E(X - EX)(Y - EY) = E(XY) - EX \cdot EY$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

## 1.6 Pearson correlation

$$\rho_{XY} = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

**Note:**  $|\text{Cov}(X, Y)| \leq \sigma_X \cdot \sigma_Y$

Hint:  $h(t) = E((X - EX) \cdot t + (Y - EY))^2 \geq 0$  for  $\forall t$ .

## 1.7 Spearman's (rank) correlation

It's defined for a pair of samples  $\mathbf{x}$  and  $\mathbf{y}$  with the same sample size  $n$ .

Let  $R(\mathbf{x})$  is the ranking of  $\mathbf{x}$ . Then Spearman's correlation is defined as

$\gamma_s(\mathbf{x}, \mathbf{y}) = \tilde{\rho}_{\mathbf{x}\mathbf{y}}$ , where  $\tilde{\rho}$  is a sample-version of Pearson correlation.